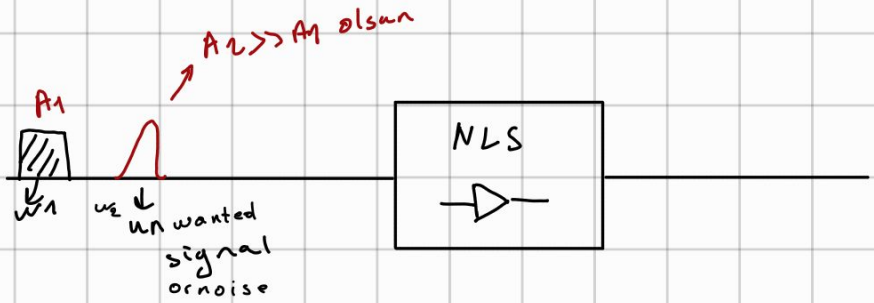
(3) Desensizitation

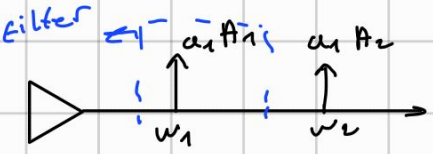


$$x(t) = \underbrace{A_1 \cos w_1 t}_{\text{desired input signal}} + \underbrace{A_2 \cos w_2 t}_{\text{unwanted signal}}$$

$$y = a_1 x + a_2 x^2 + a_3 x^3$$

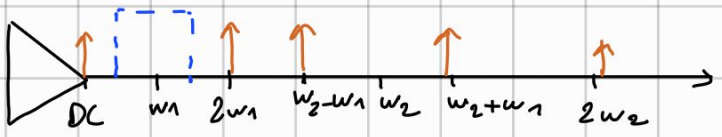
$$y(t) = a_1(A_1 \cos w_1 t + A_2 \cos w_2 t) + a_2 (A_1 \cos w_1 t + A_2 \cos w_2 t)^2 + a_3 (A_1 \cos w_1 t + A_2 \cos w_2 t)^3$$

$$= a_1 A_1 \cos w_1 t + a_1 A_2 \cos w_2 t$$



$$+ a_2 \left[A_1^2 \cos^2 w_1 t + A_2^2 \cos^2 w_2 t \right] + 2 a_2 A_1 A_2 (\cos w_1 t)(\cos w_2 t)$$

$$\cos(w_1 + w_2) t + \cos(w_2 - w_1) t$$



$$+ a_3 \left[A_1^3 \cos^3 w_1 t + A_2^3 \cos^3 w_2 t + 3 A_1 \cos w_1 t \cdot A_2^2 \cos^2 w_2 t + 3 A_2 \cos w_2 t \cdot A_1^2 \cos^2 w_1 t \right]$$

$$\cos^2 w_2 t = \frac{1 + \cos 2 w_2 t}{2}$$

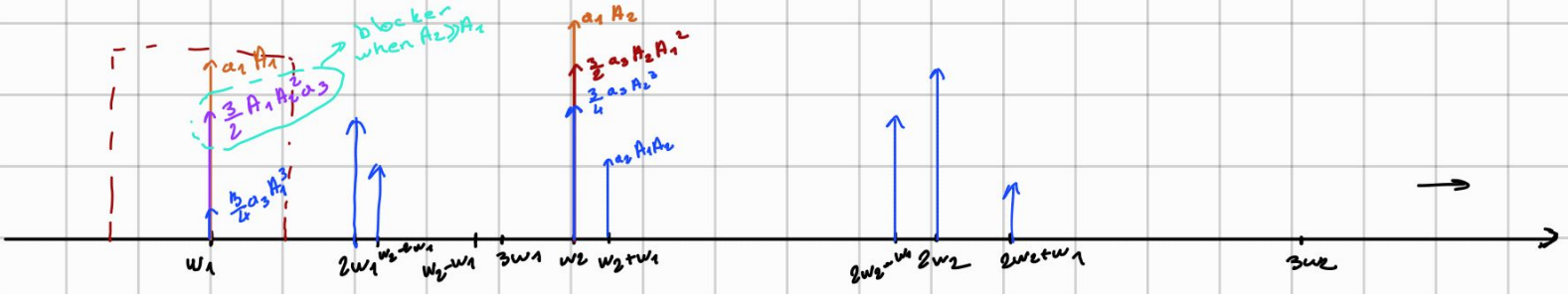
$$\frac{3}{4} a_3 A_1^3 \cos w_1 t + \frac{A_1^3}{4} \cos 3 w_1 t + \cos 3 w_2 t + \cos(2w_2 - w_1) t + \cos(2w_2 + w_1) t$$

$$\cos w_2 t + \cos w_1 t + \cos(2w_1 - w_2) t + \cos(2w_1 + w_2) t$$

$n w_1 + m w_2$

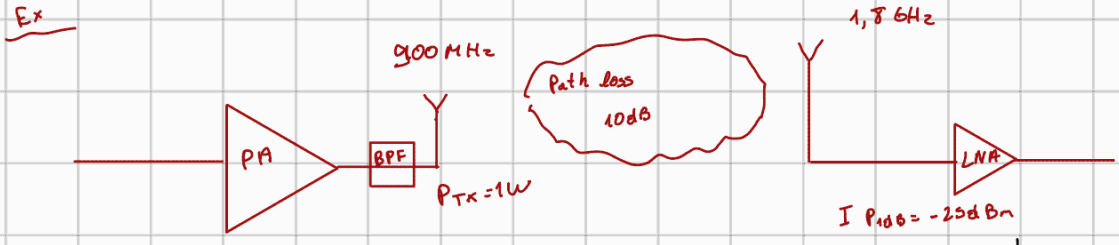
$n + |m| = 3$

$$\frac{3}{4} a_3 A_1^3 \cos w_1 t + \frac{3 A_1 A_2^2}{2} a_3 \cos w_1 t + a_1 A_1 \cos w_1 t$$

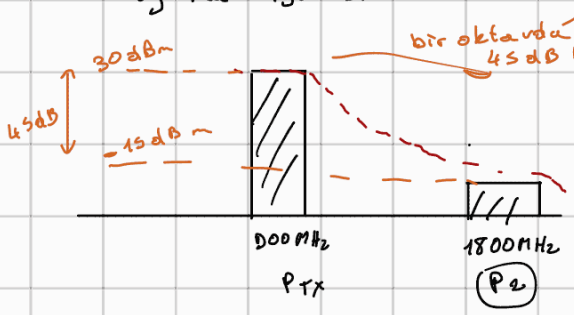


Gain: $\dfrac{(a_1 A_1 - \frac{3}{2} a_3 A_1 A_2^2)\cos\omega_1 t}{A_1 \cos\omega_1 t}$ $\Big\}$ $A_2 \uparrow\uparrow$ No signal appears at the output

Ex



Vericinin ikinci harmoniğine kadar bastırılmalıdır ki bu alıcı doğuma uğramasın



$P_2 - 10_{dB} < P_{1dB}$

$P_2 - 10 < -25\,dB$

$P_2 < -15\,dBm$

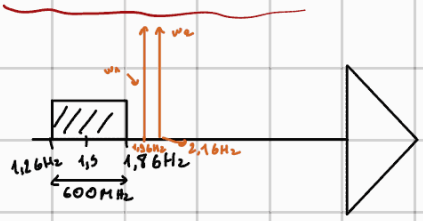
0 dBm 1 mwatt

10 dBm 10 mwat

$10 \log\left(\frac{1w}{10^{-3}}\right) = 30_{dBm}$

en az 45dB bastırılmalıdır.

④ Intermodulation



$x_1(t) = A_1 \cos\omega_1 t$

$x_2(t) = A_2 \cos\omega_2 t$

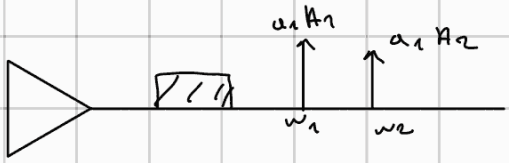
$x(t) = x_1(t) + x_2(t) = A_1\cos\omega_1 t + A_2 \cos\omega_2 t$

$g(t) = a_1 x + a_2 x^2 + a_3 x^3$

$y(t) = a_1(A_1 \cos\omega_1 t + A_2 \cos\omega_2 t)$

$+ a_2 [A_1^2 \cos^2\omega_1 t + A_2^2 \cos^2\omega_2 t + 2 A_1 A_2 \cos\omega_1 t \cos\omega_2 t]$

DC $2\omega_1$ | DC $2\omega_2$ | $\omega_2 - \omega_1$, $\omega_2 + \omega_1$



$+ a_3 [A_1^3 \cos^3\omega_1 t + A_2^3 \cos^3\omega_2 t + 3A_1 \cos\omega_1 t A_2^2 \cos^2\omega_2 t + 3A_2 \cos\omega_2 t A_1^2 \cos^2\omega_1 t]$

$\cos^2\omega_2 t = \frac{1+\cos 2\omega_2 t}{2}$ $\quad \cos^2\omega_1 t = (1+\cos 2\omega_1 t)$

ω_1, $3\omega_1$ | ω_2, $3\omega_2$ | $2\omega_2 - \omega_1$, $2\omega_2 + \omega_1$ | $2\omega_1 + \omega_2$, $2\omega_1 - \omega_2$

$2w_2 - w_1 = 2 \cdot (2{,}1\,G) - 1{,}9\,G = 2{,}3\,G$ düşmüyor

$2w_1 - w_2 = 2 \cdot (1{,}9\,G) - 2{,}1 = 1{,}7\,G \rightarrow$ In band



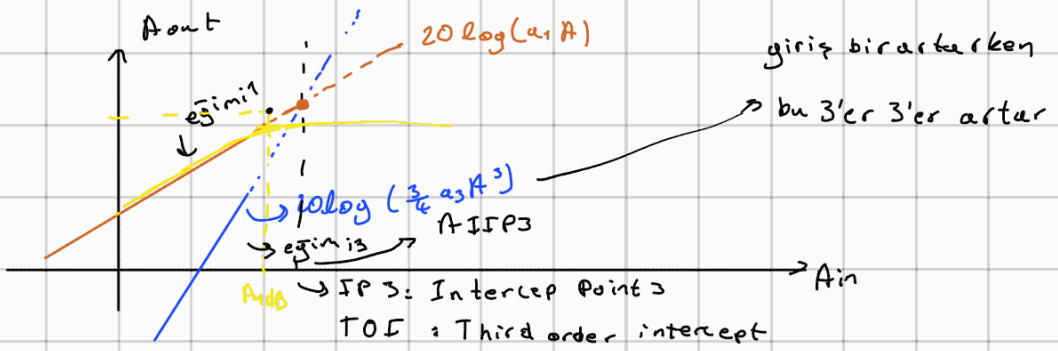
$A_{IM3} = \frac{3}{4} a_3 A_1^2 A_2 \{\cos(2w_1 - w_2)t\}$

$A_1 = A_2 = A \rightarrow A_{IM3} = \frac{3}{4} a_3 A^3$

if system linear $\rightarrow a_1 A$

Intermodulation products $\rightarrow \frac{3}{4} a_3 A^3$

$$IM3 = \frac{\frac{3}{4} a_3 A^3}{a_1 A} = \frac{3}{4} \left| \frac{a_3}{a_1} \right| A^2$$



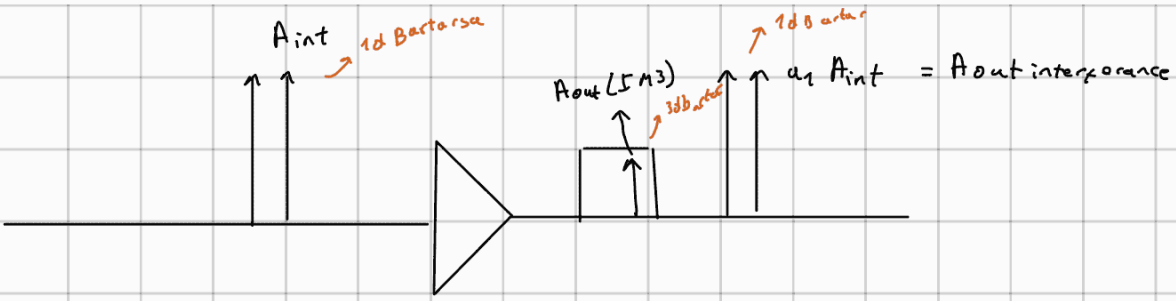
$\rightarrow A_{IIP3} \Rightarrow 20 \log(|a_1| A_{IIP3}) = 20 \log\left(\frac{3}{4} |a_3| A_{IIP3}^3\right)$

$$\boxed{A_{IIP3} = \sqrt{\frac{4}{3} \left| \frac{a_1}{a_3} \right|}}$$

$$A_{1dB} = \sqrt{0{,}145 \left| \frac{a_1}{a_3} \right|}$$

$A_{IIP3}(dB) - A_{1(dB)} = 9{,}6\,dB$

if $A_{IIP3} \uparrow \rightarrow$ more linear System

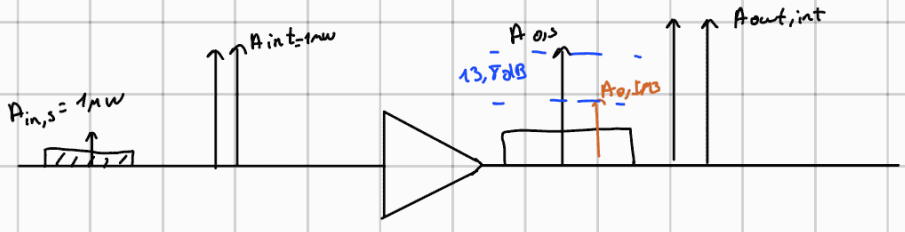


$$\frac{A_{out\ int}(\omega_1,\omega_2)}{A_{out}(IM3)} = \frac{a_1 A_{int}}{\frac{3}{4} a_3 A^3_{int}} = \frac{\frac{a_1}{\frac{3}{4} a_3 A^2_{int}}}{A^2_{IIP3}} = \frac{A^2_{IIP3}}{A^2_{int}}$$

$$A^2_{IIP3} = \frac{A_{out,int}(\omega_1,\omega_2)}{A_{out}(IM3)} \; A^2_{int}$$

$$20\log(A_{IIP3}) = 20\log(A_{int}) + \frac{1}{2}\left[20\log A_{out,in}(\omega_1,\omega_2) - 20\log A_{out}(IM3)\right]$$

Ex: $A_{in,s} = 1\mu V$, $A_{in,int} = 1mV$, $P_{IIP3} = -10\,dBm$, $Z = 50\Omega$ } Find $\frac{A_{o,s}}{A_{o,IM3}} = ?$



$$10\log\left(\frac{P_{IIP3}}{10^{-3}}\right) = -10\,dBm \quad (\text{1 skalar})$$

$$P_{IIP3} = \frac{A^2_{IIP3}}{2}$$

$$A_{IIP3} = 70\,mV$$

$$A^2_{IIP3} = \frac{A_{out,int}(\omega_1,\omega_2)}{A_{o,IM3}} \cdot A^2_{in,int} = \frac{a_1 A_{in,int}}{A_{o,IM3}} \cdot A^2_{in,int}$$

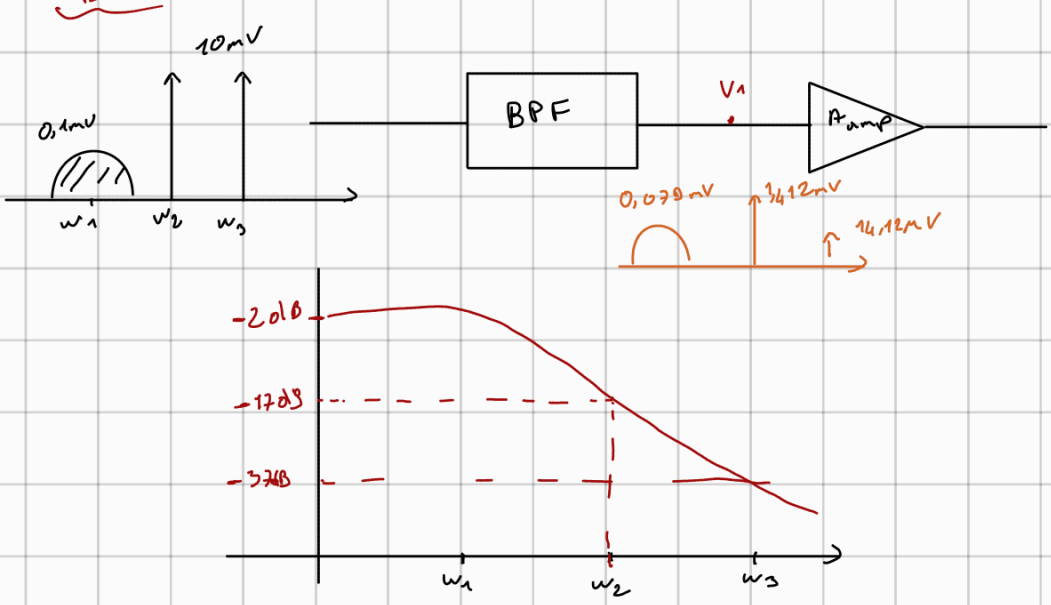
$$= \frac{a_1 A^3_{in,int}}{A_{o,im3}}$$

$$A^2_{IIP3} = \frac{a_1 A^3_{in,int}}{A_{o,im3}} \cdot \frac{A_{in,s}}{A_{in,s}} = \frac{A_{o,s}}{A_{o,IM3}} \cdot \frac{A^3_{in,int}}{A_{in,s}}$$

($A_{in,s}$ ve $A_{in,s}$ sadeleşir; sorulan $\frac{A_{o,s}}{A_{o,IM3}}$)

$$\frac{A_{o,s}}{A_{o,im3}} = A^2_{IIP3} \cdot \frac{A_{in,s}}{A^3_{in,int}} = (70mV)^2 \cdot \frac{1mV}{(1mV)^3} = 4,9 \rightarrow \approx 13,8dB$$

Ex



Compute IP3 of the amplifier such that IM3 product falling at w1 is 20 dB below the desired signal ?

→ $A_{o,IM3}$??

→ $A_{o,s}$

$$\frac{A_{o,s}}{A_{o,IM3}} \Rightarrow 20\,dB \Rightarrow \frac{A_{o,s}}{A_{o,IM3}} = 10$$

$$20\log A_{sig}(w_1) - 2dB = 20\log A_{sig_{BPF}}(w_1)$$

$$20\log \frac{A_{sig}(w_1)}{A_{sig_{BPF}}(w_2)} = 2 \Rightarrow \frac{A_{sig}(w_1)}{A_{sig_{BPF}}(w_2)} = \frac{0,1mV}{A_{sig_{BPF}}(w_1)} = 10^{2/20}$$

$$A_{sig_{BPF}}(w_1) = 0,079\,mV \quad (V_1)$$

$$20\log \underbrace{A_{int}(w_2)}_{10mV} - 17dB = 20\log A_{int_{BPF}}(w_2) \rightarrow A_{int_{BPF}}(w_2) = 1,412mV \quad (V_2)$$

$$20\log A_{int}(w_3) - 37dB = 20\log A_{int_{BPF}} \rightarrow A_{int_{BPF}}(w_3) = 14,12\mu V \quad (V_3)$$
$$0,141mV$$

$A_{o,s} = a_1 . V_1$ → $A_3(w_3) = A_2(w_2) = A$

$A_{o,IM3} = \frac{3}{4} a_3 A^3 = \frac{3}{4} a_3 A_2^2 A_3 = \frac{3}{4} a_3 V_2^2 V_3$

but $A_1(w_2) \neq A_2(w_3)$

$V_2 \neq V_3$

$$A_{o,sig}\,(dB) - A_{o,IM3}\,(dB) = 20dB$$

$$20\log|a_1 V_1| - 20\log\left|\frac{3}{4} a_3 V_2^2 V_3\right| = 20\log 10$$

$$\frac{a_1 V_1}{\frac{3}{4} a_3 V_2^2 V_3} = 10 \Rightarrow \frac{a_1}{a_3} = 26,6 \cdot 10^{-6}$$

$$A_{IIP3} = \sqrt{\frac{4}{3} \left| \frac{a_1}{a_3} \right|} = \sqrt{\frac{4}{3} \left| 26,6 \times 10^{-6} \right|} = 5,95\,mV \approx 15,5\,dBm$$

Çalışma sorusu

Ex:

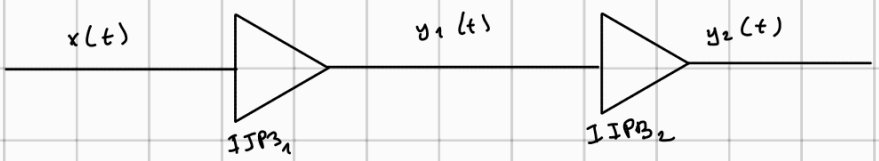
$a_1 = 10$

$A_{in,int} = 10\,mV$

$A_{IIP3} = 70\,mV$

} $IM3 = ?$, $A_{o,IM3} = ?$

Cascaded Nonlinear Stages



$$y_1(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t)$$

$$y_2(t) = b_1 y_1(t) + b_2 y_1^2(t) + b_3 y_1^3(t)$$

$$\frac{1}{A^2_{IIP3\,(total)}} = \frac{1}{A^2_{IIP3,1}} + \frac{a_1^2}{A^2_{IIP3,2}}$$

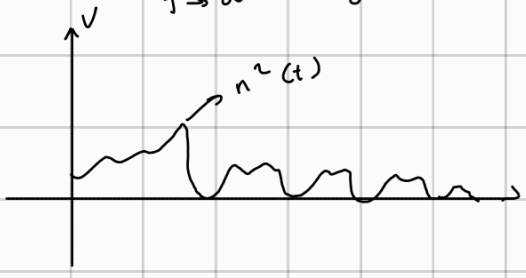
a_1^2 → gain of the first stage

$A_{IIP3,2} > A_{IIP3,1}$ → More linear system

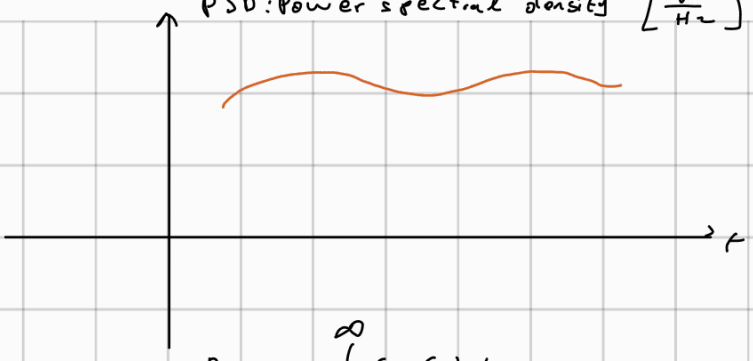
2. amplifier linearliğinin ilkinden daha iyi olmalıdır.

Noise

$$P_{ave,n} = \lim_{T \to \infty} \frac{1}{T} \int_0^T n^2(t)\,dt$$



PSD: Power spectral density $\left[\frac{V^2}{Hz}\right]$ $S_n(f)$

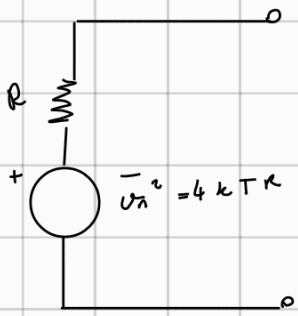


$$P_{ave} = \int_0^\infty S_n(f)\,df$$

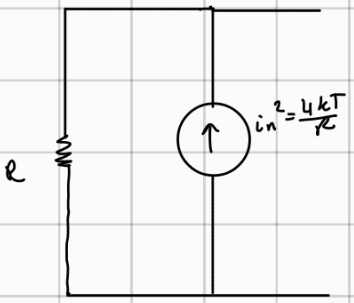
Device Noise



Resistor Noise:



veya

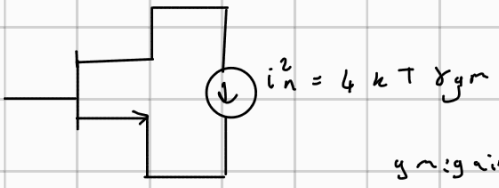


$1,3 \times 10^{-23}$ J/K

k: Boltzman sabiti

T = (kelvin) sıcaklık

MOS Transistor Noise



g_m: gain

γ : noise coefficient

